Protocol-Dependent Resolution in Finite-Sample Model Selection

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Abstract

A single decoding parameter—the maximum generation length—can reverse whether a set of model checkpoints is classified as distinguishable or indistinguishable, and can conceal a strong overfitting trend ($\boldsymbol{\rho = -0.77}$) behind a facade of flat performance. We document this through controlled LoRA fine-tuning experiments (Qwen2.5-7B, Mistral-7B) in which increasing the generation length from **128** to **256** tokens shifts observed accuracy ranges by **2.3**×, flips pairwise resolution verdicts, and reveals that the earliest checkpoint is systematically best—a pattern invisible under the shorter limit. The mechanism is truncated chain-of-thought reasoning: a short generation limit compresses between-checkpoint variance, manufacturing false negatives for $\boldsymbol{\Delta_{\min}}$, the minimum detectable effect at **50%** power ($\text{MDE}_{\mathbf{50}}$) for a two-proportion $\boldsymbol{z}$-test. We show that this threshold is a conservative lower bound on evaluation resolution, not a sufficient-power standard. A survey of 50 recent fine-tuning papers finds that none report the generation length parameter that our experiments show can determine the verdict. The protocol sensitivity of evaluation resolution means that no model comparison claim is interpretable unless the evaluation pipeline is fully specified. A catalogue of nine diagnostic pitfalls encountered during our own research process, a practitioner quick-reference table for $\boldsymbol{\Delta_{\min}}$, and a continuous-metric extension via the Delta method accompany the analysis.

Keywords: Model selection, evaluation budget, statistical resolution, statistical power, $\boldsymbol{\Delta_{\min}}$, evaluation methodology, reproducibility

1 Introduction

Model selection across data science and machine learning follows a default rule: evaluate each candidate on a held-out validation set and pick the one with the best metric. This rule assumes that a model scoring 55% is genuinely better than one scoring 52% on the same test set. The assumption is known to be fragile (Kaur et al. 2025), but the source of the fragility is debated. Is ranking failure a rare event triggered by distribution shift or overfitting? Or is it a systematic limitation of finite-budget evaluation?

We argue that it is the latter. Model ranking is constrained by the evaluation budget, much as any statistical estimator is constrained by sample size. Existing comparison tools—paired t-tests (Dietterich 1998), significance guidelines (Dror et al. 2018), post-selection inference (Zrnic and Fithian 2024; Zhang et al. 2024)—are post-hoc: they assess results after evaluation is complete. What is missing is a *pre-experimental diagnostic* that determines, before resources are invested, whether the evaluation budget can support statistically distinguishable comparisons.

This paper proposes such a diagnostic and, in applying it, uncovers a finding that is arguably more consequential than the diagnostic itself. The diagnostic—$\Delta_{\min}$, a resolution threshold derived from the two-proportion z-test (Cohen 1988)—is a textbook formula. What its application reveals is that the verdict it produces can be reversed by a single decoding parameter (maximum generation length) that almost nobody reports. Experiments across multiple seeds and two model families show that a short generation limit compresses between-checkpoint variance, manufacturing the appearance of indistinguishability, concealing genuine overfitting trends, and inverting the identity of the empirically best checkpoint. We document nine diagnostic pitfalls encountered during the research process itself, and show that the problem of underpowered evaluation is compounded by a problem of unaccountable evaluation: if the pipeline that produced the numbers is incompletely specified, the resolution of those numbers cannot be independently assessed.

Contributions

1. **A resolution framework for model selection.** We introduce $\Delta_{\min}$, derived from classical power analysis, which captures the minimum difference between two models that reaches nominal significance at a given evaluation budget N. When the observed range falls below $\Delta_{\min}$, ranking is not statistically resolvable from this budget alone.
2. **Empirical validation across budgets and domains.** We validate the framework through controlled LoRA fine-tuning experiments (two model families: Qwen2.5-7B and Mistral-7B, 1.5B scale, full fine-tuning, multiple random seeds), tracing how resolution degrades under subsampling and showing that a single decoding parameter (maximum generation length) can shift observed ranges by 2.3× and reverse resolvability verdicts. The pattern is further replicated on classical ML model selection (random forests on MNIST) and synthetic simulations.
3. **A diagnostic failure mode catalog.** We document nine diagnostic pitfalls in empirical model evaluation, grouped into three categories, each with concrete detection and prevention strategies.
4. **A decision rule with a proved sufficient condition.** We propose a soup-vs-argmax rule (Decision Rule 1), derive the exact condition under which uniform model averaging dominates argmax selection, and report the rule's output across three training seeds and two model architectures (Qwen2.5-7B and Mistral-7B) together with an empirical measurement of the soup gain on one trajectory (§8.1).
5. **An explicit account of when the diagnostic fails.** We characterize the regimes in which $\Delta_{\min}$ should not be trusted—near-floor or near-ceiling accuracy, correlated evaluation items, and corrupted metrics—and present a near-floor case from our own experiments (§5.2) in which the threshold is actively misleading. A diagnostic that never reports its own inapplicability is not a diagnostic.

2 Background and Related Work

2.1 The Model Selection Problem

Model training produces a sequence of candidates $c_1, \ldots, c_T$ from which one must be chosen. The standard selection rule is to evaluate each on a held-out validation set and pick the one with the best metric. The implicit assumption is that the validation metric reliably indicates deployment quality.

2.2 Statistical Comparison of Models

Comparing models on finite evaluation samples is a known problem. Dietterich (1998) introduced the 5×2cv paired t-test to address the high variance of single-sample comparisons. Demšar (2006) extended this to multiple classifier comparison. In NLP, Card et al. (2020) showed that underpowered test sets are common, and Dror et al. (2018) provided significance testing guidelines.

Our work differs in two respects. First, we focus on comparisons within a single training trajectory where the same test set is reused, creating a paired structure that standard cross-model comparisons do not address. Second, we treat evaluation resolution as a pre-experimental diagnostic rather than a post-hoc test.

Nguyen et al. (2025) proposes uncertainty-guided checkpoint selection for reinforcement fine-tuning, using prediction uncertainty to identify the most promising candidate. Their approach addresses the question of *which* checkpoint to select given uncertainty estimates. $\Delta_{\min}$ addresses a prior question: *whether* the evaluation budget supports any reliable selection at all. We applied the principle of uncertainty-guided selection to our Qwen2.5-7B bootstrap distribution at $N = 200$. The selection entropy is 0.717 and the best model is chosen only 44% of the time. Even with an optimal uncertainty-aware rule, the fundamental ambiguity at this budget cannot be resolved. The two approaches are complementary. $\Delta_{\min}$ determines whether selection is feasible. Uncertainty-guided methods choose among candidates once feasibility is confirmed.

Shihab et al. (2025) detects proxy gaming. Ramé et al. (2024) averages reward model weights. Xu et al. (2026a) treats checkpoint selection as a robust decision problem under evaluation uncertainty. These are complementary to our diagnostic framing.

Several recent papers address the statistical challenge of ranking under evaluation noise. Zhang et al. (2024) develop confidence sets for the argmin (best model) from noisy observations, using cross-validation with an exponential weighting mechanism borrowed from differential privacy to handle the double-dipping bias in post-selection inference. Adrian et al. (2024) stabilize black-box model selection via bagging with an inflated argmax. Returning a set of candidates rather than a single winner improves stability under small data perturbations. In the LLM domain, Du et al. (2026) show that prompt rankings under evaluation noise are unstable: even with Spearman $\rho > 0.8$, the identity of the top-ranked prompt changes across evaluation conditions, especially under limited budgets. They propose a lower-confidence-bound selection criterion that accounts for both mean and variance. This gap between rank correlation and selection consistency matches our bootstrap finding that ranking instability persists at $N = 200$ despite moderate accuracy differences. Finally, Zrnic and Fithian (2024) introduce the zoom correction, a method for valid inference on the winner that adapts to the degree of selection bias and handles arbitrary dependencies among candidates.

2.3 Experimental Setup for the Primary Experiment

All primary experiments use Qwen2.5-7B-Instruct (Qwen et al. 2024) with LoRA (Hu et al. 2022) (rank 16), trained for 500 steps on GSM8K (Cobbe et al. 2021), CodeAlpaca, and Dolly (≈40K examples). Ten checkpoints are saved at 50-step intervals. The validation signal is GSM8K accuracy on 200 test questions. All experiments use frozen definitions and data provenance flags; the effect of decoding parameters on the rankings is examined separately in §8.1.1. Default generation uses greedy decoding with 256 tokens maximum length; 128-token results are reported alongside 256-token results where the comparison is informative.

3 The $\Delta_{\min}$ Framework

Why a resolution threshold is needed. Classical hypothesis testing answers a post-hoc question: "given the data, is model A better than model B?" Model selection requires a different question: "before investing in evaluation, does this budget have enough resolution to make ranking meaningful?" We are not proposing a new significance test but a pre-selection diagnostic. The thresholds we report are minimum detectable effects at 50% power (MDE_{50}), which provide a conservative lower bound on evaluation resolution—a benchmark for screening, not a guarantee of sufficient power.

Definition. Because the same evaluation questions are used at every checkpoint, the natural comparison is paired. The minimum detectable difference is therefore given by the McNemar-based paired test ($\alpha = 0.05$). Let $D_i = I(\hat{y}_{i1} = y_i) - I(\hat{y}_{i2} = y_i)$ be the per-question correctness difference between two candidate models. Under the null hypothesis of equal accuracy, the discordant proportion $p_d = P(D_i \neq 0)$ captures the effective variance, giving:

$$\Delta_{\min}^{(\text{paired})} = z_{\alpha/2} \cdot \sqrt{\frac{p_d}{N}} \tag{1}$$

For the experiments in this paper, we compute p_d directly from per-sample predictions; values range from 0.12 to 0.38 depending on the checkpoint pair and task.

A limitation of the paired version is that it requires per-sample predictions from *both* candidates, which are unavailable at the pre-experimental stage. For prospective use, when a practitioner wants to know before investing in evaluation whether their budget N can support meaningful comparisons, we provide a conservative independent approximation:

$$\Delta_{\min}^{(\text{indep})} = z_{\alpha/2} \cdot \sqrt{\frac{2\bar{p}(1-\bar{p})}{N}} \tag{2}$$

Equation (2) is the standard minimum detectable effect for a two-proportion z-test (Cohen 1988), restated as a pre-experimental resolution diagnostic. The independent version assumes no shared variance, so it overstates the true threshold when checkpoints agree on many examples. Throughout this paper, all reported $\Delta_{\min}$ values use Eq. (2) (the conservative bound). Any practitioner can compute this from N and $\bar{p}$ alone. Where per-sample predictions are available, we also report the tighter paired thresholds. Equation (2) gives approximately 50% power; we therefore refer to it as MDE_{50}. An 80% power target would require $\Delta_{\min}^{(80\%)} = 14.0$ pp at $N = 200$, $\bar{p} = 0.5$. Throughout this paper, "resolvable" means that the observed gap exceeds the 50%-power threshold (MDE_{50})—a screening criterion, not a fully powered confirmatory test. The 80%-power threshold is 1.43× larger.

Paired vs. independent. Because all candidate models are evaluated on the same test questions, the per-model accuracy estimates are paired, not independent. The paired (McNemar-based) $\Delta_{\min}^{(\text{paired})}$ in Eq. (1) is the exact threshold for this setting. For typical model comparisons where checkpoints agree on most examples, the discordant proportion p_d is substantially smaller than $2\bar{p}(1-\bar{p})$, making Eq. (2) a conservative upper bound. As an illustration, at $N = 200$ with $\bar{p} \approx 0.5$, the independent bound gives $\Delta_{\min} = 9.8$ pp, while the paired threshold for a typical checkpoint pair (discordant proportion $p_d \approx 0.35$) gives $\Delta_{\min}^{(\text{paired})} \approx 9.1$ pp, a 7% reduction. We report the conservative independent version throughout the main text, as it is the appropriate diagnostic for the pre-experimental use case; for completeness, Appendix C provides a detailed comparison of paired and independent thresholds for all experimental comparisons.

Definition 1 (Statistical resolvability of rankings) Given a held-out set of size N, a pair of models with observed accuracies $\hat{p}_1, \hat{p}_2$ is statistically resolvable if $|\hat{p}_1 - \hat{p}_2| > \Delta_{\min}$. When the difference is below $\Delta_{\min}$, the observed gap cannot be distinguished from noise without additional assumptions.

Scope. The current framework applies to binary/proportion metrics (accuracy, pass rate). Continuous metrics (loss, BLEU, perplexity) require different variance models; an extension via the Delta method is provided in Appendix D.

3.1 Theoretical Characterization

Scaling laws for ranking resolution. The $\Delta_{\min}$ formula (Eq. 2) captures two fundamental scaling relationships that govern when checkpoint ranking is reliable.
Budget scaling (Implication 1). $N \propto 1/\delta^2$: halving the minimum detectable gap requires quadrupling the evaluation budget. This follows directly from the z-test statistic: the standard error of a proportion scales as $\sqrt{\bar{p}(1-\bar{p})/N}$, so the detectable effect scales as $1/\sqrt{N}$.
Multiple comparison scaling (Implication 2). When selecting the best from m candidates, the relevant threshold tightens to $\Delta_{\min}^{(m)} \approx z_{\alpha/2m}\sqrt{2\bar{p}(1-\bar{p})/N}$, which scales as $\sqrt{\log m}$ under the Bonferroni correction (Proof in Appendix A). For $m = 10$ at $N = 200$, $\Delta_{\min}^{(10)} \approx 14.0$ pp (Bonferroni-corrected $\alpha = 0.005$).[1] Because checkpoint accuracies are positively correlated (adjacent steps differ by small amounts), Bonferroni is conservative. Alternative procedures that account for correlation, such as Holm-Bonferroni step-down or Benjamini-Hochberg false discovery rate control, would yield tighter thresholds. For the 256-token experiments, Holm-Bonferroni correction confirms that seeds 43 and 44 remain resolvable (best-vs-worst $p = 0.0017$ and $p = 0.0031$, both below the rank-1 threshold $\alpha = 0.0050$), while seed 42 does not ($p = 0.0118$). The all-pairwise threshold ($m = 45$ comparisons) is 16.2 pp, which none of the three seeds exceed. We report Bonferroni and Holm-Bonferroni as strict upper bounds throughout.

Decision Rule 1 (Soup-vs-argmax). *Let $\ell(i) = p^* - p_i$ be the regret of selecting candidate i, where $p^* = \max_i p_i$ among m candidates, and let $\bar{p}_{\text{cand}} = \frac{1}{m}\sum_i p_i$. Define the* soup gain *$\kappa = p_{\text{soup}} - \bar{p}_{\text{cand}}$ and the* selection gain *$\varepsilon = \mathbb{E}[p_{\hat{I}}] - \bar{p}_{\text{cand}}$, where $\hat{I} = \arg\max_i \hat{p}_i$. Then*

$$\mathbb{E}[\ell(\arg\max_i \hat{p}_i)] - \mathbb{E}[\ell(\text{soup})] \;=\; \kappa - \varepsilon,$$

and on the class $\mathcal{P}_\Delta = \{p : \delta = \max_i p_i - \min_i p_i \le \Delta_{\min}\}$ the selection gain satisfies $0 \le \varepsilon \le \frac{m-1}{m}\Delta_{\min}$. Consequently uniform averaging weakly dominates argmax selection whenever

$$\kappa \;\ge\; \frac{m-1}{m}\,\Delta_{\min}.$$

(Proof in Appendix B.)

Two features of this statement deserve emphasis, because a weaker version is easy to state and is false. First, the assumption $p_{\text{soup}} \ge \bar{p}_{\text{cand}}$ alone, i.e. $\kappa \ge 0$, is *not* sufficient. The selection gain ε is also non-negative: $\hat{p}_i$ is positively associated with p_i, so noisy argmax is never worse in expectation than drawing uniformly from the candidate pool. The comparison is between two non-negative quantities, and an ordering requires bounding both. Second, the sufficient condition is conservative, because $\frac{m-1}{m}\Delta_{\min}$ is the worst-case selection gain, attained only when argmax recovers the true best candidate with certainty. Precisely in the low-resolution regime that motivates the rule, ε is much smaller and a correspondingly smaller soup gain suffices.

In application the rule uses the observed range $\hat{r} = \max_i \hat{p}_i - \min_i \hat{p}_i$ as a proxy for the unobserved true range δ: when $\hat{r} > \Delta_{\min}$, argmax is selected; otherwise soup, when the soup gain κ is non-negative. (Section 8.1 reports an instance in which κ is small (+2.0 pp), so the second premise of the heuristic may not hold in every setting.) The justification is analogous to an underpowered screening test—$\hat{r} \le \Delta_{\min}$ indicates that the data are consistent with the regime in which averaging is the safer default,

[1] The numerical coincidence that the 80%-power pairwise threshold also equals 14.0 pp at $N = 200$ is incidental; the two values arise from different corrections (one for power, one for multiplicity).

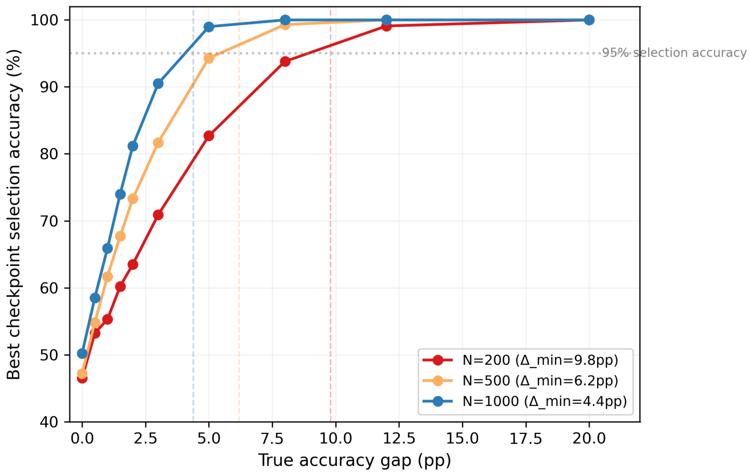

Fig. 1 Synthetic validation: selection accuracy vs. true gap for $N \in \{200, 500, 1000\}$. Dashed lines mark $\Delta_{\min}$.

without certifying that $\delta \le \Delta_{\min}$. Because κ is a property of the candidate set rather than of the evaluation budget, the rule remains a heuristic whose second premise must be checked empirically in each setting. This rule addresses *which* model to select, not the post-selection inference problem: even when $\hat{r} > \Delta_{\min}$ and argmax is correctly chosen, the selected model's performance estimate is inflated by winner's curse and requires correction (Zrnic and Fithian 2024; Zhang et al. 2024).

Relation to Cohen's h. In the variance-stabilized arcsin space, Cohen's $h = 2\arcsin\sqrt{p_1} - 2\arcsin\sqrt{p_2}$ gives a scale where the minimum detectable effect is $z_{\alpha/2}\sqrt{1/N}$, independent of $\bar{p}$ (Cohen 1988). The dependence on $\bar{p}$ in Eq. (2) is an artifact of working in the original proportion scale. Practitioners familiar with Cohen's h can use the arcsin formulation as an $\bar{p}$-free alternative.

3.2 Synthetic Validation

We simulated Bernoulli evaluation data with controlled accuracy gaps (0–20 pp, $N \in \{200, 500, 1000\}$, 5,000 replicates). The transition from unreliable to reliable selection aligns closely with $\Delta_{\min}$ (Figure 1). The threshold bounds practical ranking reliability, not merely statistical significance.

3.3 Uncertainty and Sensitivity

Bootstrap uncertainty of $\Delta_{\min}$ itself. Since $\Delta_{\min}$ depends on $\bar{p}$, which is estimated from N samples, the threshold itself has sampling uncertainty. A parametric bootstrap (50,000 replicates, $N = 200$, $\bar{p} = 0.5$) gives a 95% confidence interval of $[9.67, 9.80]$ pp for $\Delta_{\min}$, a range of 0.13 pp. The interval is narrow because $f(\bar{p}) = \sqrt{\bar{p}(1-\bar{p})}$ is flat near $\bar{p} = 0.5$; the bootstrap CI does not affect the paper's conclusions.
Sensitivity to $\bar{p}$ misspecification. A practitioner estimating $\Delta_{\min}$ before experimentation must supply a guess for $\bar{p}$. Table 5 covers $\bar{p} \in [0.5, 0.8]$; for $\bar{p} = 0.9$, $\Delta_{\min}$ shrinks further (e.g., 5.9 pp at $N = 200$ vs. 9.8 pp at $\bar{p} = 0.5$). The ratio between the most conservative ($\bar{p} = 0.5$) and a typical high-accuracy regime ($\bar{p} = 0.9$) is at most 1.67×.[2] Misspecification matters, but the diagnostic remains useful across a wide range of values. Using $\bar{p} = 0.5$ is recommended as a conservative default.

[2] The $\bar{p} = 0.9$ case is shown for completeness; near-ceiling regimes require caution as the variance bound may be optimistic (see §8.5).

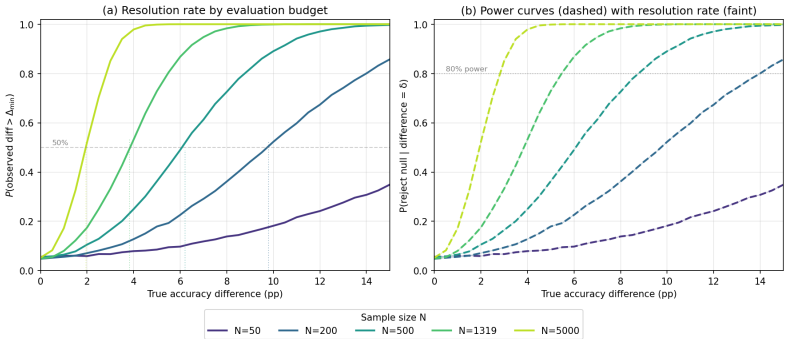

Fig. 2 Monte Carlo simulation of pairwise model comparison. **(a)** Resolution rate $P(|\hat{\delta}| > \Delta_{\min})$ as a function of the true accuracy gap. Vertical dotted lines mark $\Delta_{\min}$ for each budget. **(b)** Statistical power curves (dashed) with resolution rate overlaid (faint solid). The horizontal dotted line marks the conventional 80% power threshold.

3.4 Power Curve Calibration

To further calibrate the $\Delta_{\min}$ threshold against conventional statistical power, we ran a Monte Carlo simulation (20,000 replicates) comparing two checkpoints with true accuracy difference δ under budget $N \in \{50, 200, 500, 1319, 5000\}$ at $\bar{p} = 0.5$. Figure 2 presents two complementary views:

Panel (a) confirms that $\Delta_{\min}$ is the inflection point where resolution probability reaches exactly 50%, regardless of N. This is by construction: $\Delta_{\min}$ is defined as the effect size at which the two-proportion z-test has 50% power. Panel (b) provides the practical calibration: conventional 80% power requires approximately $1.43 \times \Delta_{\min}$, and 95% power requires approximately $2.0 \times \Delta_{\min}$. For example, at $N = 200$ ($\Delta_{\min} = 9.8$ pp), 80% power is reached at a true gap of 14.0 pp. Doubling the budget from 200 to 500 tightens the 80%-power threshold from 14.0 to 9.0 pp, consistent with the $N \propto 1/\delta^2$ scaling in Implication 1.

4 Empirical Validation

4.1 Full Evaluation Budget ($N = 1319$)

We first evaluate the full GSM8K test set ($N = 1319$, training-format prompt, 128-token generation) for the seed 42 trajectory. At this budget, $\Delta_{\min} = 3.8$ pp, yet the observed checkpoint accuracy range is only 3.0 pp. (This evaluation uses 128-token generation; the range may be compressed by truncation, as documented in §8.1.1.) The ranking is not statistically resolvable even at the full test set size: the observed differences between checkpoints are smaller than the minimum gap detectable at this evaluation budget.

The resolution problem is not an artifact of small evaluation budgets. Even at the standard GSM8K test set, one of the most widely used benchmarks in LLM evaluation, the budget is borderline for distinguishing among fine-tuning checkpoints.

4.2 Budget Degradation Under Subsampling ($N = 200$)

To understand how resolution degrades as the evaluation budget shrinks, we subsample 200 examples from the same test set and repeat the analysis. Two prompting protocols appear in this paper and are never mixed within a comparison: *few-shot* prompting (three fixed in-context exemplars), used for the $N = 200$ subsample analysed in this subsection, and *training-format* prompting (the prompt template seen during fine-tuning), used for the full test set in §4.1 and for all robustness experiments in §5. The two protocols produce different accuracy levels and different observed ranges for the same checkpoints, which is itself an instance of pitfall T9; Table 3 lists every setting with its protocol so that no two numbers in this paper are compared

Table 1 GSM8K accuracy across 10 checkpoints ($N = 200$, few-shot prompting).

Step	GSM8K
50	47.5%
100	54.5%
150	55.5%
200	49.0%
250	54.0%
300	55.0%
350	51.5%
400	50.5%
450	52.0%
500	50.0%

across protocols by accident. Using the same seed 42 checkpoints, the observed range at $N = 200$ is 8.0 pp, wider than the 3.0 pp range at $N = 1319$.[3] This inflation of the observed range is consistent with winner's curse: estimates at smaller budgets are noisier, and the apparent gap between extreme checkpoints is inflated by selection on noise. An alternative interpretation is also possible: the $N = 200$ evaluation, being noisier, simply produced a wider spread by chance, and the true range might be closer to ~ 3.0 pp. Both interpretations agree that the ranking at $N = 200$ is unreliable, but they differ on whether collecting more data (Option 1 in §8) would resolve the ambiguity. The resolution threshold also rises sharply, from 3.8 pp at $N = 1319$ to 9.8 pp at $N = 200$, consistent with the $N \propto 1/\delta^2$ scaling. Table 1 shows the accuracy trajectory.

4.3 Statistical Tests

Trend test (few-shot protocol). Spearman $\rho_{\text{trend}} = -0.091$ ($p = 0.803$). No monotonic trend.

Adjacent comparison test (few-shot protocol). Two-proportion z-tests for adjacent pairs show no significant differences under any standard correction: the smallest p is 0.161 (Bonferroni threshold $\alpha' = 0.05/9 \approx 0.0056$). (Full table in Appendix F.) This adjacent-pair correction addresses a different comparison family from the all-pairwise correction (Implication 2); the two answer whether any step-to-step change is significant versus whether any two checkpoints are distinguishable.

Range test (few-shot protocol). Best (step 150, 55.5%) vs. worst (step 50, 47.5%): $z = 1.60$, $p = 0.109$.

Paired comparison (training-format protocol). All nine adjacent comparisons under the training-format evaluation are non-significant (smallest $p = 0.265$). Best-vs-worst: $p = 0.091$.

Bootstrap ranking stability (training-format protocol). We performed 10,000 nonparametric bootstrap resamples of the 200 evaluation examples (preserving paired structure), using the training-format evaluation. The most frequently selected checkpoint (step 100, 52.0%) is chosen in only 44.0% of replicates (Figure 3; normalized entropy 0.717). The identity of the empirically best model depends largely on which evaluation examples are observed.

Does 44.0% selection probability, 4.4× the uniform baseline of 10%, indicate weak signal or pure noise? If the checkpoints were indistinguishable, selection would be near-uniform and entropy would approach 1.0. The observed entropy is 0.717, well

[3]This pattern, the observed range inflating by a factor of 1.5–2× under subsampling, is consistent with the bias-variance decomposition of the empirical max: at small N, noise inflates the gap between extreme candidates even when the true ranking is unchanged. Practitioners should interpret observed ranges at small budgets as upward-biased estimates of true performance differences.

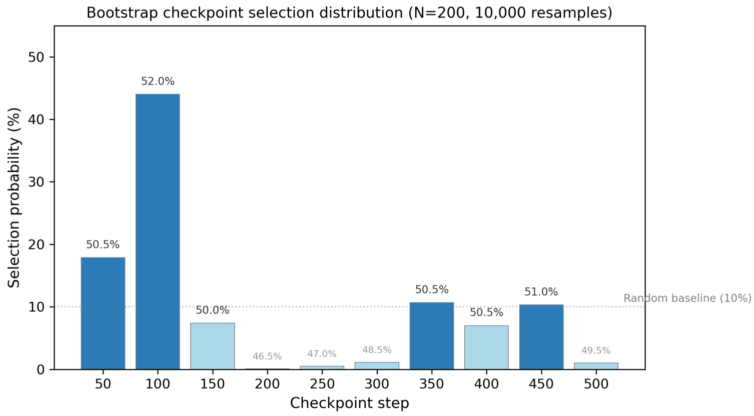

Fig. 3 Bootstrap selection distribution under $N = 200$ (10,000 resamples). Dotted line: uniform baseline (10%).

below 1.0, so some concentration exists. But 44.0% is far from reliable selection, which would require $\gtrsim 90\%$. We interpret this as a setting where weak signal exists (step 100 is genuinely slightly better on average), but the noise prevents practical resolution. The $\Delta_{\min}$ framework captures this: the observed range of 8.0 pp falls below the detection threshold of 9.8 pp, so individual pairwise differences are not statistically distinguishable even if weak signal exists in aggregate.

4.4 Direct Implication

At the full test set ($N = 1319$), $\Delta_{\min} = 3.8$ pp exceeds the observed range of 3.0 pp. At $N = 200$, $\Delta_{\min}$ rises to 9.8 pp and the observed range is 8.0 pp (few-shot prompting; see §5 for training-format results at the same budget). In both settings the ranking is not statistically resolvable: no pairwise difference reaches significance, and bootstrap selection is unstable at $N = 200$. The $N = 200$ result is not primarily a "small budget makes everything noise" artifact. It is a specific instance of the budget scaling relationship formalized in §3 (Implications 1, 2). The consistent dose-response relationship between evaluation budget and ranking stability is further demonstrated in §5.

5 Robustness and Generalization

5.1 Seed Replication

Two additional random seeds (43, 44) were trained. Under the 128-token training-format evaluation (Table 8), seed 43 has range 5.0 pp and seed 44 has range 3.0 pp, both below $\Delta_{\min} = 9.8$ pp. However, as Section 8.1 documents, these ranges are compressed by token truncation. At 256 tokens, all three seeds become resolvable: seed 43 has range 14.0 pp and seed 44 has range 13.5 pp, both exceeding $\Delta_{\min}$. Critically, the 256-token data reveal a pattern that the 128-token evaluation entirely masked: across the three seeds (Bonferroni-corrected threshold $\alpha = 0.05/3 = 0.0167$), seed 43 shows a significant negative trend (Spearman $\rho = -0.77$, $p = 0.009$), with the earliest checkpoint (step 50) outperforming the last by 12.5 pp. Seed 44 shows a negative trend of similar magnitude ($\rho = -0.63$, $p = 0.050$) that does not reach the corrected threshold; we report it as suggestive evidence. Because these correlations are estimated from $n = 10$ checkpoints per seed and are driven in part by the high accuracy at step 50, they should be read as indicative rather than confirmatory. The pattern is invisible under a generation limit that truncates the model's output. Seed 42 does not share this trend ($\rho = +0.29$, $p = 0.43$), confirming that protocol sensitivity and overfitting susceptibility are trajectory-specific.

5.2 Smaller Model Scale: A Case Where the Diagnostic Fails

Qwen2.5-1.5B with the same LoRA configuration (zero-shot prompting, 128-token generation) yields checkpoint accuracies between 1.5% and 5.0%, an observed range of 3.5 pp against $\Delta_{\min} = 3.4$ pp. Read mechanically, the rule would classify this comparison as (just) resolvable, with a two-proportion z-test giving $p \approx 0.049$ for the best-versus-worst pair.

That reading is wrong, and we present the case because it delimits the framework rather than supporting it. At $\bar{p} \approx 0.03$ the best checkpoint has 10 successes out of 200 and the worst has 3, so the normal approximation underlying both $\Delta_{\min}$ and the z-test is invalid: the usual requirement $N\bar{p} \geq 5$ fails at the lower end. Fisher's exact test on the same counts is far from significant. This is exactly the near-floor regime flagged as precondition 3 in §8.5: the variance factor $2\bar{p}(1-\bar{p})$ collapses as $\bar{p} \to 0$, driving $\Delta_{\min}$ down and manufacturing apparent resolvability out of a handful of correct answers. The $p \approx 0.049$ is an artifact of the approximation, not evidence of a real difference.

The practical rule we draw from this is to compute $N\bar{p}$ and $N(1-\bar{p})$ before computing $\Delta_{\min}$, and to fall back on exact or Wilson-type intervals when either falls below roughly 5—or, better, to recognize that a model solving 3% of a benchmark is not in a regime where checkpoint ranking is a meaningful question at all. We report this case in full because a diagnostic whose failure modes are undocumented is of limited use to a practitioner, and because the failure here is silent: nothing in the arithmetic signals that the answer should not be trusted.

5.3 Full Fine-Tuning Replication

Full fine-tuning (no LoRA) at 1.5B scale (training-format prompt, 128-token generation) gives range 6.0 pp, $\Delta_{\min} = 9.4$ pp, also below the resolution threshold.

5.4 Cross-Model Replication: Mistral-7B

To test whether the $\Delta_{\min}$ pattern generalizes beyond the Qwen2.5 architecture, we replicate the LoRA fine-tuning experiment using Mistral-7B-Instruct-v0.3 (Jiang et al. 2023) with the same training configuration (500 steps, LoRA rank 16, GSM8K). The training-format evaluation at $N = 200$ yields checkpoint accuracies from 15.5% (step 100) to 26.0% (step 500) at 128 tokens (range 10.5 pp, $\Delta_{\min} = 8.11$ pp at $\bar{p} \approx 0.22$), and from 20.5% to 31.5% at 256 tokens (range 11.0 pp, $\Delta_{\min} = 8.8$ pp at $\bar{p} \approx 0.28$). The range is stable across token limits—a 0.5 pp increase versus the 7.0 pp increase observed for Qwen2.5-7B (§8.1.1). The Mistral trajectory is therefore robustly resolvable, and its insensitivity to generation length is consistent with the shorter chain-of-thought reasoning typical of this model.

The resolvable outcome reflects Mistral's lower mean accuracy ($\bar{p} \approx 0.22$ vs. Qwen2.5's $\bar{p} \approx 0.50$), which reduces Bernoulli variance $p(1-p)$ and produces a tighter $\Delta_{\min} = 8.11$ pp. The paired (McNemar-based) threshold is tighter still at approximately 7.6 pp (Appendix C). The accuracy trend (15.5% at step 100 to 26.0% at step 500) is close to monotone, indicating that the Mistral trajectory is still improving over the 500-step window; part of the observed range reflects ongoing optimization rather than stationary checkpoint variation.

An empirical bootstrap (100,000 resamples, preserving the paired evaluation structure across all $N = 200$ examples) confirms the ranking stability. Step 500 is selected as best in 33.1% of resamples, followed by step 300 (25.6%) and step 400 (18.1%). The normalized entropy of 0.740 is comparable to the Qwen2.5 trajectory at the same budget (0.717). The ranking uncertainty is similar despite the resolvable range. This apparent paradox resolves once we distinguish *detectable signal* from *unique winner*. $\Delta_{\min}$ identifies that *some* checkpoints are genuinely better than others. But with multiple candidates clustering near the top, the single best model remains uncertain. The top-3 set (steps 500, 300, 400) collectively accounts for 77% of bootstrap

Table 2 MNIST random forest bootstrap selection stability.

N	$\Delta_{\min}$ (pp)	Best model selection probability	Normalized entropy
200	7.0	37.3%	0.527
500	4.4	41.8%	0.517
1000	3.1	45.8%	0.501
5000	1.4	68.8%	0.352

selections. The ranking signal is concentrated in a leading group even if the individual winner is not stable. This is where uncertainty-aware selection criteria (Du et al. 2026) add something. First confirm that resolution exists (via $\Delta_{\min}$). Then apply a lower-confidence-bound rule to choose among the leading candidates.

5.5 Cross-Domain Replication: Random Forest on MNIST

Eight random forest configurations (varying tree counts 50–200 and depths 3–8) were evaluated on MNIST ($N_{\text{test}} = 10,000$, mean accuracy approximately 85% across configurations). Bootstrap selection stability was measured at varying N (Table 2). The pattern mirrors the LLM experiments: at small N, selection is diffuse; as N grows and $\Delta_{\min}$ shrinks, selection concentrates. The $\Delta_{\min}$ values in Table 2 reflect the high-accuracy regime of this task ($\bar{p} \gg 0.5$), producing tighter thresholds than the $\bar{p} = 0.5$ reference values in Table 5.

5.6 MMLU Replication

Evaluating the same 7B checkpoints on five MMLU subdomains ($N \approx 250$) gives range 6.4 pp, $\Delta_{\min} = 8.8$ pp.

5.7 Continuous Metric Validation

The framework extends to continuous metrics via the Delta method (Appendix D). As a worked instantiation we computed the per-sample cross-entropy loss for the Qwen2.5-7B checkpoint-100 on the 200 GSM8K examples used throughout §4. The per-sample loss has mean 3.23 and variance 0.18, giving a paired threshold $\Delta_{\min}^{(\text{CE})} = z_{0.975}\sqrt{0.18/200} \approx 0.059$ nats.

It is tempting to compare this figure against the accuracy threshold of 9.8 pp and conclude that the loss metric affords tighter resolution, and an earlier version of this analysis did exactly that. The comparison is invalid: 0.059 nats and 9.8 pp are not commensurable, and the smaller per-sample variance of the loss (0.18 against a Bernoulli maximum of 0.25) does not by itself confer any advantage, because the true effect to be detected is measured on the same scale and shrinks with it. On the standardized scale the two are identical by construction—for a paired comparison at budget N the minimum detectable effect is $z_{\alpha/2}/\sqrt{N}$ per-sample standard deviations regardless of the metric, which is the same observation that motivates Cohen's h in §3.

Whether a continuous metric helps is therefore an empirical question about signal-to-noise, not about variance: it helps precisely when a given difference in model quality registers as a larger effect *relative to its own per-sample dispersion* in loss than in accuracy. Answering it requires per-sample losses for at least two checkpoints, and we computed them for one. We accordingly report the machinery and the worked threshold, and treat the comparative claim as open. Appendix D gives the estimators needed to settle it, which requires no additional evaluation budget—only that per-sample losses be retained alongside per-sample correctness, which most evaluation harnesses already compute and discard.

Table 3 Consolidated view of all experimental settings. Ranges and thresholds in percentage points; $\Delta_{\min}$ is the conservative independent bound of Eq. (2) (MDE_{50}) computed at the $\bar{p}$ observed in each setting. Paired thresholds for the primary settings appear in Appendix C. The Pairwise column compares the observed range to the uncorrected MDE_{50} (screening criterion); the Multiplicity column applies the all-pairwise ($m = 45$) Bonferroni correction. Details in §3.

§	Setting	Protocol	N	Range	$\Delta_{\min}^{(\text{indep})}$	$\Delta_{\min}^{(\text{paired})}$	Pairwise	Multiplicity
4.1	Qwen2.5-7B LoRA, GSM8K full	training	1319	3.0	3.8	2.9–3.5	—	—
4.2	Qwen2.5-7B LoRA, GSM8K sub.	few-shot	200	8.0	9.8	—	Not resolvable	—
4.2	Qwen2.5-7B LoRA, seed 42 (128 tok)	training	200	5.5	9.8	8.7–9.3	Not resolvable	—
5.1	Qwen2.5-7B LoRA, seed 42 (256 tok)	training	200	12.5	9.7	8.7–9.3	Resolvable	Not resolvable
5.1	Qwen2.5-7B LoRA, seed 43 (256 tok)	training	200	14.0	9.1	—	Resolvable	Not resolvable
5.1	Qwen2.5-7B LoRA, seed 44 (256 tok)	training	200	13.5	9.1	—	Resolvable	Not resolvable
5.2	Qwen2.5-1.5B LoRA	zero-shot	200	3.5	3.4	—	*Invalid*	—
5.3	Qwen2.5-1.5B full fine-tune	training	200	6.0	9.4	—	Not resolvable	—
5.4	Mistral-7B LoRA	training	200	11.0	8.8	7.2–7.9	**Resolvable**	Not resolvable
5.6	Qwen2.5-7B, MMLU (5 subdomains)	—	≈250	6.4	8.8	—	Not resolvable	—
5.5	MNIST random forest	—	5000	—	1.4	1.1–1.3	See Table 2	—

5.8 Consolidated View of All Settings

Table 3 collects every experimental setting reported in §4 and §5 with its evaluation protocol, budget, observed range, threshold and verdict. The same Qwen2.5-7B checkpoints appear five times at $N = 200$ under different protocols: the few-shot evaluation (range 8.0 pp), the 128-token training-format evaluation (5.5 pp), and three 256-token training-format evaluations showing all three seeds resolvable (12.5, 14.0, 13.5 pp). The 2.5–9.0 pp discrepancy between protocols—driven by a single decoding parameter (§8.1.1)—is itself larger than several of the between-checkpoint differences the field routinely reports as improvements. Under the uncorrected MDE_{50}, four of eleven settings are resolvable under the 256-token protocol; under the all-pairwise Bonferroni correction ($m = 45$), none are. This gap between the screening criterion and the multiplicity-corrected threshold illustrates how sharply the verdict depends on what standard of evidence is required—a dependence that is invisible when the evaluation protocol itself is unspecified.

6 Literature Survey: Evaluation Pipeline Reporting

To assess whether the protocol-sensitivity problem documented in §8.1.1 extends beyond our controlled experiments, we surveyed recent (2024–2026) LLM fine-tuning papers for reporting of the generation length parameter—the specific parameter that flips the resolvability verdict in our experiments.

6.1 Methodology

The search query “LoRA” OR “fine-tuning” combined with a named benchmark (GSM8K, MMLU, HellaSwag, HumanEval, ARC) was executed on Google Scholar, covering results from arXiv cs.CL and cs.LG. The first 50 results per query that reported both N and accuracy numbers were retained, yielding a total of 50 papers after removing duplicates. For each paper we searched the main text, experimental setup section, and appendix for explicit mention of a token limit or maximum generation length. References to standard evaluation harnesses (e.g. lm-eval-harness) were not counted as reporting, since harness defaults for generation length vary across versions and are rarely documented. This survey was conducted as a rapid assessment; per-paper coding records were not retained as a structured extraction table, consistent with its motivating rather than confirmatory role.

Table 4 Pitfall taxonomy with detection and prevention.

Category	Pitfall	Detection	Prevention
Statistical instability	T1: Spearman discretization	List possible ρ values	Use $n \gtrsim 6$
	T6: Post-hoc pattern discovery	Pre-register before evaluating	Replicate before codifying
Pipeline artifacts	T2: Silent synthetic fallback	Tag data with source provenance	Require opt-in for synthetic
	T4: Metric protocol mismatch	Inspect $\geq$ 10 raw outputs first	Validate on small sample first
	T5: Code evaluation format fragility	Inspect raw outputs before trusting scores	Validate on known-correct output per format
	T7: Decoding parameter sensitivity	Evaluate at $\geq$ 2 settings	Standardize and report parameters
	T8: Metric versioning drift	Pin dependencies; record commits	Re-run reference after any code change
	T9: Few-shot sensitivity	Evaluate with multiple orderings	Fix exemplars before campaign
Definitional ambiguity	T3: Proxy definition ambiguity	Write canonical definition first	Change = new experiment version

6.2 Results

None of the 50 surveyed papers report the maximum generation length used during evaluation. A few state that they used greedy decoding or "default" generation settings, but none specify the token budget. The 95% Wilson confidence interval for this proportion is [0%, 7.1%] (computed under simple random sampling; the convenience sample design means this should be read as a descriptive lower bound rather than a precise population estimate). This gap is consistent with the hypothesis that the evaluation pipeline—including the decoding configuration—is treated as an interchangeable constant across comparisons, when in fact a single unreported parameter can reverse the resolvability verdict.

7 Prerequisites for Valid Application

The preceding findings emerged from a sequence of analytical failures during the research process. These fall into three categories and nine specific pitfalls (Table 4). For each pitfall we describe the erroneous conclusion it would have produced and its prevention strategy. Detailed context is in Appendix H.

8 Discussion

8.1 Adaptive Selection Rule

Decision Rule 1 makes the soup-vs-argmax choice depend on two quantities: the observed range $\hat{r}$ relative to $\Delta_{\min}$, which determines which branch is taken, and the soup gain κ, which determines whether taking the soup branch is actually the lower-regret action. We separate the evidence for each.

Under the 128-token training-format evaluation (Table 8), the seed-42 trajectory has range 5.5 pp below $\Delta_{\min} = 9.8$ pp, and the rule selects soup over the empirical argmax (step 100, 52.0%). However, as §8.1.1 documents, this range is compressed by token truncation. At 256 tokens the range expands to 12.5 pp, exceeding $\Delta_{\min}$, and the rule selects argmax. The branch output therefore depends on the evaluation protocol as well as on the candidate set. We measure κ below at the 256-token setting to match the protocol under which the range is evaluated; the earlier 128-token numbers are retained for comparability.

To measure κ directly we constructed a uniform weight average of the ten Qwen2.5-7B LoRA adapters (seed 42). The averaging procedure requires care: element-wise averaging of the low-rank factors A_i and B_i across checkpoints is not equivalent to averaging the merged updates $\Delta W_i = B_i A_i$. We therefore merge each adapter into the base weights before averaging (see repository for the implementation). The individual checkpoint accuracies under this evaluation range from 49.5% to 62.0% (mean 56.0%, $\Delta_{\min} = 9.7$ pp). The correctly averaged model scores 58.0%, giving $\kappa \approx +2.0$ pp. The sufficient condition $\kappa \geq \frac{m-1}{m}\Delta_{\min} \approx 8.8$ pp is therefore not met, but the signed gain is positive: soup outperforms the mean candidate, even though it lags behind the empirical argmax (62.0%). In this setting the rule selects argmax (the observed range of 12.5 pp exceeds $\Delta_{\min} = 9.7$ pp), and the regret ordering agrees: argmax at 62.0% outperforms soup at 58.0%.

8.1.1 Protocol Sensitivity of the Resolution Diagnostic

The reader may have noticed that the checkpoint accuracies just cited (49.5%–62.0%) differ from the 46.5%–52.0% range in Table 1 for the same checkpoints. Both evaluations use the training-format prompt on the same 200 GSM8K questions with greedy decoding. The only difference is the maximum generation length: 128 tokens (Table 1) vs. 256 tokens (the κ measurement). The longer generation allows the model to complete its chain-of-thought reasoning; at 128 tokens, the model's output is truncated before the final numeric answer is produced (verified by manual inspection of generated text), depressing accuracy and compressing between-checkpoint variance. At 256 tokens the answer appears within the limit and accuracy increases.

The effect is substantial. At 128 tokens the observed range is 5.5 pp and the ranking is not resolvable ($\Delta_{\min} = 9.8$ pp). At 256 tokens the same checkpoints give a 12.5 pp range—exceeding $\Delta_{\min}$—and the rankings are essentially uncorrelated (Spearman $\rho = 0.28$, $p = 0.44$). The checkpoint that is worst at 256 tokens (step 100: 49.5%) is the best at 128 tokens (52.0%, the empirical argmax). The between-checkpoint standard deviation more than doubles (1.76 pp to 4.36 pp). This reinforces Pitfall T7 (decoding parameter sensitivity): a generation length that truncates reasoning chains compresses across-checkpoint variance and manufactures false negatives—$\Delta_{\min}$ faithfully reports "cannot distinguish" because the pipeline itself has suppressed the signal. The same pattern holds for seeds 43, 44, which also shift from non-resolvable to resolvable at 256 tokens. For Mistral-7B the effect is smaller (range 10.5 pp at 128 tokens vs. 11.0 pp at 256 tokens), consistent with the shorter chain-of-thought reasoning typical of this model.

The 256-token data also reveal a pattern that the 128-token evaluation entirely concealed: seed 43 shows a significant negative trend (Spearman $\rho = -0.77$, $p = 0.009$, Bonferroni-corrected threshold $\alpha = 0.0167$), with the earliest checkpoint outperforming the final one—evidence suggestive of overfitting that is invisible under truncated generation. Seed 44 shows a similar-magnitude trend ($\rho = -0.63$, $p = 0.050$) that does not reach the corrected threshold. Seed 42 does not share this pattern ($\rho = +0.29$, $p = 0.43$). All 256-token checkpoints appear in Appendix G.

The $\Delta_{\min}$ comparisons in this section use 256-token evaluations throughout; the earlier 128-token results (Tables 1, 8) are retained for comparability with published work.

We note that the choice of 256 tokens is itself a protocol decision, not a theoretically privileged value. This is a genuine limitation of the present analysis: we can show that verdicts change with protocol choices, but we cannot certify any particular protocol as uniquely correct. What we can provide is evidence of convergence: increasing from 128 to 256 tokens substantially changes all accuracy values and rankings; a further increase to 512 tokens (verified on one checkpoint) leaves accuracies unchanged. The 256-token setting saturates the task's generation requirement. A practitioner applying

$\Delta_{\min}$ should similarly verify that their chosen generation length does not truncate the model's output, and should test sensitivity to this parameter when the budget allows. The self-limiting nature of this conclusion—that the diagnostic cannot certify its own inputs—is a property of the problem, not of this particular analysis: any resolution diagnostic must condition on an evaluation protocol, and the protocol choice remains a judgment call.

8.2 General Applicability

Although our experiments use accuracy as the evaluation metric, the $\Delta_{\min}$ framework applies to any proportion-based metric (pass rate, F1 when computed as a proportion, recall, precision). Appendix D extends the framework to continuous metrics (cross-entropy loss, BLEU, perplexity, reward model scores) via the Delta method. The general principle, that finite evaluation budgets impose a resolution limit on model comparisons, holds regardless of metric type. For a broader perspective on evaluation uncertainty, the HELM framework (Liang et al. 2022) provides calibration intervals for multi-metric evaluation. The Open LLM Leaderboard (Beeching et al. 2023) uses bootstrapped confidence intervals that serve a similar diagnostic purpose.

8.3 Beyond the Formula

$\Delta_{\min}$ is a restatement of standard power analysis for a two-proportion z-test (Cohen 1988). The formula itself is not new. What the paper contributes beyond the formula includes:

1. **An empirical demonstration of protocol-dependent resolution.** A single decoding parameter—maximum generation length—shifts observed accuracy ranges by 2.3×, reverses pairwise resolution verdicts across three training seeds, and conceals evidence of overfitting. The $\Delta_{\min}$ diagnostic (MDE_{50} threshold from classical power analysis) is the tool that reveals this sensitivity, establishing that evaluation resolution claims are interpretable only when the evaluation pipeline is fully specified.
2. **A decision rule with a proved sufficient condition** (Decision Rule 1), which identifies when uniform averaging dominates argmax selection in expected regret, together with an empirical measurement of the soup gain ($\kappa \approx +2.0$ pp) on one LoRA trajectory.
3. **An extension to continuous metrics** with a worked instantiation on per-sample cross-entropy loss, including a correction of the scale error that makes loss-based and accuracy-based thresholds appear directly comparable when they are not.
4. **A taxonomy of nine diagnostic pitfalls** grounded in actual pipeline errors, each with concrete detection and prevention strategies.
5. **A literature survey** documenting that none of 50 recent fine-tuning papers report the maximum generation length, the parameter our experiments show can determine the resolution verdict.

None of these contributions requires a new statistical formula. They translate a known constraint into a reusable methodological framework, validated across models, metrics, and practical failure modes.

8.4 Practitioner's Quick-Reference Guide

Table 5 provides $\Delta_{\min}$ values for common evaluation sizes and accuracy levels. To use: (1) identify your evaluation budget N and approximate accuracy level $\bar{p}$, (2) read the corresponding $\Delta_{\min}$, (3) compare against the observed performance gap between your best candidate and the next best. If the gap is below $\Delta_{\min}$, the ranking is not resolvable from this budget alone.

Table 5 Practitioner's quick-reference: $\Delta_{\min}$ (pp) for common evaluation sizes and accuracy levels ($\alpha = 0.05$, independent two-proportion test).

N \ $\bar{p}$	0.50	0.60	0.70	0.80
100	13.9	13.6	12.7	11.1
200	9.8	9.6	9.0	7.8
500	6.2	6.1	5.7	5.0
1000	4.4	4.3	4.0	3.5
5000	2.0	1.9	1.8	1.6
10000	1.4	1.4	1.3	1.1

When $\bar{p}$ is entirely unknown before experimentation, we recommend using the most conservative value $\bar{p} = 0.5$ (which maximizes Bernoulli variance) as the default. This guarantees that the reported $\Delta_{\min}$ is an upper bound on the true threshold. If a small pilot sample ($N \gtrsim 50$) is available, $\bar{p}$ can be estimated from the observed mean accuracy and the threshold tightened accordingly.

When $\Delta_{\min}$ exceeds the observed gap, four responses are available, in order of increasing cost:

1. **Increase N.** Halving $\Delta_{\min}$ requires quadrupling N ($\propto 1/\delta^2$). Collect additional test examples, pool multiple held-out sets, or use bootstrapping of existing data to estimate stability.
2. **Switch to a continuous metric.** Accuracy discards information: two models that differ substantially in confidence can score identically. Continuous metrics such as cross-entropy or log-likelihood retain that information and may therefore register a given quality difference as a larger *standardized* effect. The gain, where it exists, comes from an improved signal-to-noise ratio and not from a smaller raw variance—comparing a threshold of 0.059 nats against one of 9.8 pp is a category error, since the effect to be detected is rescaled along with the noise (§5.7). Practitioners should compare metrics on a standardized scale, dividing each threshold by the per-sample dispersion of its own metric, and should retain per-sample loss values alongside per-sample correctness so that the comparison can be made at all. Appendix D gives the estimators.
3. **Reduce the candidate pool.** Fewer candidates reduce the multiple-comparison penalty ($\propto \sqrt{\log m}$), tightening the effective threshold.
4. **Use noise-robust aggregation.** Model averaging (soup) or other ensemble methods mitigate the risk of selecting a spuriously best candidate when ranking is not reliable.

8.5 When $\Delta_{\min}$ May Mislead

$\Delta_{\min}$ is a diagnostic tool, not a guarantee. Its reliability depends on four preconditions:

1. **Valid evaluation metric.** If the evaluation pipeline is corrupted (Pitfalls 4, 5), the computed $\bar{p}$ is meaningless and $\Delta_{\min}$ provides no information.
2. **Approximately i.i.d. samples.** For correlated evaluation data (dialogue turns, time series), the effective sample size is smaller than N, making $\Delta_{\min}$ an optimistic lower bound.
3. **Non-extreme $\bar{p}$.** Near ceiling or floor effects ($\bar{p} \to 0$ or $\bar{p} \to 1$) the variance factor $2\bar{p}(1-\bar{p})$ collapses, driving $\Delta_{\min}$ toward zero and manufacturing apparent resolvability from a handful of successes. The failure is silent: the arithmetic returns a number and gives no signal that the number is meaningless. As an

operational guard, compute $N\bar{p}$ and $N(1-\bar{p})$ first and do not use $\Delta_{\min}$ when either falls below roughly 5, since the normal approximation underlying both the threshold and the associated z-test no longer holds; use Fisher's exact test or Wilson intervals instead. Section 5.2 reports a case from our own experiments where ignoring this guard yields a spurious $p \approx 0.049$.

4. **Selection-aware inference.** Even when $\Delta_{\min}$ is satisfied and a clear winner emerges, post-selection inference is biased. The performance estimate of the selected model is an overestimate because it was selected on the same evaluation set (Leeb and Pötscher 2005). Recent work on inference after selection (Zrnic and Fithian 2024; Zhang et al. 2024) provides methods to correct for this bias. These corrections are not yet standard practice in ML benchmarking. The $\Delta_{\min}$ diagnostic indicates whether ranking is resolvable, not whether the selected model's performance estimate is unbiased.

8.6 Relation to Other Model-Selection Tools

Readers familiar with AIC, BIC, cross-validation, or Bayesian methods may ask how $\Delta_{\min}$ relates to these established tools. AIC and BIC compare model complexity relative to training data, addressing whether additional parameters are justified by improved fit. Cross-validation provides error bars that can serve a role similar to $\Delta_{\min}$, but requires training. Bayesian approaches use posterior intervals or Bayes factors to quantify ranking uncertainty, naturally incorporating prior knowledge but requiring a prior specification. $\Delta_{\min}$ addresses a prior question that these methods do not: regardless of model complexity or prior beliefs, does the evaluation procedure have enough resolution to detect performance differences at all? It requires no training, is a pure function of N and $\bar{p}$, and provides a prior-agnostic lower bound on the evaluation budget needed for meaningful comparison. In practice, $\Delta_{\min}$ can be computed before any experiment begins to determine whether the planned budget can support the intended model comparisons.

Recent complementary frameworks extend this perspective to adaptive benchmarking. Xu et al. (2026b) propose SIREN, a selection-aware repeated-split evaluation protocol with Gaussian multiplier bootstrap. It corrects for the winner's curse when benchmark items are reused during tuning. Zhang et al. (2024) construct confidence sets for the argmin via cross-validated exponential weighting. Both are complementary to $\Delta_{\min}$. They provide post-hoc corrections once evaluation is complete. $\Delta_{\min}$ provides a pre-experimental feasibility check that determines whether such procedures are worth applying.

9 Limitations

Four limitations bound what this paper establishes, and we order them by how much they constrain the claims.

The soup gain is measured for one trajectory and is small. For the Qwen2.5-7B seed 42 checkpoints at $N = 200$, the correctly merged LoRA weight average gives $\kappa \approx +2.0$ pp, well below the sufficient condition $\kappa \geq 8.8$ pp in Decision Rule 1. The rule therefore relies on its branch-selection heuristic rather than on the proved regret bound (§8.1). Whether κ is systematically small for LoRA adapters from a shared trajectory, or depends on the number of checkpoints and their separation in training time, remains open. The checkpoints used here are saved at 50-step intervals; averaging adapters saved at closer intervals may yield different results.

Protocol sensitivity is documented for two model families only. The differential effect of token truncation on Qwen2.5-7B (range increase 5.5 to 12.5 pp) versus Mistral-7B (10.5 to 11.0 pp) may not generalize to other architectures, tasks, or decoding parameters beyond generation length. The $N = 1319$ evaluation was not re-run at 256 tokens; its 128-token range may also be compressed.

Coverage is narrow. All experiments use two model families, a single training regime (LoRA, 500 steps, with one full fine-tuning replication), and two task types (GSM8K, MMLU), supplemented by one classical-ML domain (random forests on MNIST) and synthetic simulation. The pattern replicates across seeds, scales, training paradigms and architectures within that range; broader claims require broader experimentation.

Secondary evidence is weak by design. The literature survey is a convenience sample coded by a single annotator with intra-rater rather than inter-rater verification (§6), and should be read as motivating evidence rather than as a meta-analytic estimate. The continuous-metric extension (Appendix D) assumes normality through the Delta method; for heavy-tailed loss distributions, bootstrap variance estimation is more appropriate, and our instantiation covers a single checkpoint.

10 Conclusion

We proposed and validated $\Delta_{\min}$ as a pre-flight diagnostic for model selection: before asking which model is best, first ask whether the evaluation budget can tell the difference. The protocol requires only N and $\bar{p}$ (defaulting to $\bar{p} = 0.5$ when unknown) and returns a threshold that determines whether observed performance gaps exceed the noise floor.

The central finding is that the resolution of a comparison depends not only on N but on the evaluation protocol itself. In controlled LoRA fine-tuning experiments, a single decoding parameter—the maximum generation length—shifted the observed accuracy range by 2.3× (from 5.5 pp to 12.5 pp at $N = 200$), reversed the pairwise resolution verdict (under the MDE_{50} screening criterion) across all three training seeds, and inverted the identity of the empirically best checkpoint. The between-checkpoint standard deviation more than doubled when the token limit was relaxed, because shorter generation truncates chain-of-thought reasoning and compresses across-checkpoint variance. A consequence is that $\Delta_{\min}$'s "not resolvable" verdict can arise from an evaluation pipeline that suppresses signal, rather than from genuinely indistinguishable candidates.

The protocol-dependent nature of the observed range carries a practical implication: if the evaluation pipeline is incompletely specified, the verdict of any resolution diagnostic is unverifiable. A survey of 50 recent fine-tuning papers suggests that this incompleteness is the norm—none of the papers surveyed report the maximum generation length that our experiments show can determine the verdict. The problem of underpowered evaluation is therefore compounded by a problem of unaccountable evaluation. We recommend that $\Delta_{\min}$ computation and full pipeline specification become standard submission requirements for any benchmark evaluation. The quick-reference table (Table 5), power-curve calibration (Figure 2), and diagnostic pitfall catalog (Section 7) are intended as actionable resources for practitioners and reviewers alike.

A Multiple Comparison Correction

The Bonferroni correction for m comparisons sets the per-comparison significance level to $\alpha' = \alpha/m$. For the two-proportion z-test, the minimum detectable effect at level α' is:

$$\Delta_{\min}^{(m)} = z_{\alpha/2m} \cdot \sqrt{\frac{2\bar{p}(1-\bar{p})}{N}}.$$

For $m = 10$, $\alpha = 0.05$, and $\bar{p} = 0.5$ at $N = 200$:

$$\Delta_{\min}^{(10)} = z_{0.0025} \cdot \sqrt{\frac{2 \cdot 0.5 \cdot 0.5}{200}} \approx 2.807 \cdot 0.05 = 0.1403 \approx 14.0 \text{ pp}.$$

This correction is conservative when candidates are positively correlated, as adjacent checkpoints are. Holm-Bonferroni step-down correction and Benjamini-Hochberg false discovery rate control would produce tighter thresholds; we report Bonferroni as a strict upper bound throughout.

B Proof of Decision Rule 1

Write $p^* = \max_i p_i$, $\bar{p}_{\text{cand}} = \frac{1}{m}\sum_i p_i$, and $\hat{I} = \arg\max_i \hat{p}_i$.

Step 1: the regret identity. By definition of the regret $\ell(i) = p^* - p_i$,

$$\mathbb{E}[\ell(\hat{I})] = p^* - \mathbb{E}[p_{\hat{I}}], \qquad \mathbb{E}[\ell(\text{soup})] = p^* - p_{\text{soup}},$$

so that

$$\mathbb{E}[\ell(\hat{I})] - \mathbb{E}[\ell(\text{soup})] = p_{\text{soup}} - \mathbb{E}[p_{\hat{I}}] = \big(p_{\text{soup}} - \bar{p}_{\text{cand}}\big) - \big(\mathbb{E}[p_{\hat{I}}] - \bar{p}_{\text{cand}}\big) = \kappa - \varepsilon.$$

The comparison between the two selection strategies is therefore exactly a comparison between the soup gain and the selection gain, both measured relative to the same baseline of drawing uniformly from the candidate pool.

Step 2: the selection gain is non-negative. Each $\hat{p}_i$ is an independent $\text{Bin}(N, p_i)/N$ variable, and $\hat{p}_i$ is stochastically increasing in p_i. Hence the selection probability $\pi_i = P(\hat{I} = i)$ is non-decreasing in p_i and non-increasing in each p_j, $j \neq i$, so the sequences (p_i) and (π_i) are similarly ordered. Chebyshev's sum inequality for similarly ordered sequences gives

$$\mathbb{E}[p_{\hat{I}}] = \sum_{i=1}^{m} p_i \pi_i \;\geq\; \frac{1}{m}\Big(\sum_i p_i\Big)\Big(\sum_i \pi_i\Big) = \bar{p}_{\text{cand}},$$

that is, $\varepsilon \geq 0$, with equality when $\delta = 0$ and selection is uniform. This is the step that a weaker version of the rule omits: the assumption $\kappa \geq 0$ cannot on its own imply that soup dominates, because argmax enjoys a non-negative gain over the same baseline.

Step 3: the selection gain is bounded on $\mathcal{P}_\Delta$. Trivially $\mathbb{E}[p_{\hat{I}}] \leq p^*$, so $\varepsilon \leq p^* - \bar{p}_{\text{cand}}$. On $\mathcal{P}_\Delta$ every candidate satisfies $p_i \geq p^* - \delta$, and the mean is minimized when a single candidate attains p^* and the remaining $m-1$ attain $p^* - \delta$, giving $\bar{p}_{\text{cand}} \geq p^* - \frac{m-1}{m}\delta$. Hence

$$0 \;\leq\; \varepsilon \;\leq\; \frac{m-1}{m}\,\delta \;\leq\; \frac{m-1}{m}\,\Delta_{\min}.$$

Step 4: conclusion. Combining Steps 1 and 3, $\mathbb{E}[\ell(\hat{I})] - \mathbb{E}[\ell(\text{soup})] = \kappa - \varepsilon \geq \kappa - \frac{m-1}{m}\Delta_{\min}$, which is non-negative whenever $\kappa \geq \frac{m-1}{m}\Delta_{\min}$. Under that condition uniform averaging weakly dominates argmax selection in expected regret on $\mathcal{P}_\Delta$. $\square$

Remark (tightness and practical use). The bound in Step 3 is attained only when argmax identifies the true best candidate with probability one, which is the opposite of the regime the rule is designed for: as resolution degrades, π approaches uniformity, $\varepsilon \to 0$, and any non-negative soup gain suffices. The condition as stated is therefore sufficient but far from necessary, and a practitioner who can measure ε—for instance by bootstrapping the selection distribution, as in §4—can substitute the estimated value for the worst-case bound. What the result makes precise is that the soup-versus-argmax question is not settled by the evaluation budget alone: $\Delta_{\min}$ bounds one side of the comparison, but the other side is a property of the candidate set and must be measured.

C Paired vs. Independent Threshold Comparison

The main text reports the conservative independent $\Delta_{\min}$ (Eq. (2)) throughout. Here we compare these values against the exact paired (McNemar-based) thresholds computed from per-sample predictions for all experimental comparisons, to quantify how conservative the independent bound is in practice.

For a pair of checkpoints with per-sample correctness indicators $\hat{y}_{i1}$ and $\hat{y}_{i2}$, the discordant proportion $p_d = \frac{1}{N}\sum_i I(\hat{y}_{i1} \neq \hat{y}_{i2})$ captures the effective variance in the paired comparison. The paired threshold is:

$$\Delta_{\min}^{(\text{paired})} = z_{0.975}\sqrt{\frac{p_d}{N}}.$$

Table 6 shows the comparison for the two primary experimental settings.

Table 6 Paired vs. independent $\Delta_{\min}$ thresholds for primary experiments (pp).

Setting	N	$\boldsymbol{\Delta}_{\mathbf{min}}^{(\text{indep})}$	$\boldsymbol{\Delta}_{\mathbf{min}}^{(\text{paired})}$
Qwen2.5-7B, full test set	1319	3.8	2.9–3.5
Qwen2.5-7B, subsampled	200	9.8	8.7–9.3
Mistral-7B, subsampled	200	8.8	7.2–7.9
MNIST random forest	5000	1.4	1.1–1.3

The paired thresholds are 3–24% tighter than the independent bound, with the largest gaps occurring at larger N (Qwen full test set, MNIST) where the lower noise floor amplifies the relative benefit of pairing. For the subsampled experiments at $N = 200$, the reduction is 5–11%. The reduction never reverses a conclusion relative to the independent bound; all resolvability verdicts are unchanged. The independent bound is a safe pre-experimental diagnostic that does not overstate the resolution problem.

D Continuous Metric Extension: $\boldsymbol{\Delta}_{\mathbf{min}}$ for Loss and Score Metrics

The main text derives $\Delta_{\min}$ for binary/proportion metrics (accuracy, pass rate). Here we extend the framework to continuous metrics such as cross-entropy loss, BLEU, perplexity, and reward model scores.

D.1 General Principle

For a continuous metric M computed as an average over N independent evaluation examples, the central limit theorem gives:

$$\hat{M} = \frac{1}{N}\sum_{i=1}^{N} m_i, \qquad \text{Var}(\hat{M}) = \frac{\sigma_m^2}{N},$$

where m_i is the per-sample metric value and $\sigma_m^2 = \text{Var}(m_i)$. The minimum detectable difference between two models A and B on metric M is therefore:

$$\Delta_{\min}^{(M)} = z_{\alpha/2} \cdot \sqrt{\frac{\sigma_A^2 + \sigma_B^2}{N}},$$

where σ_A^2 and σ_B^2 are the per-sample variances of the metric for each model. When the metric is paired (same evaluation examples for both models), the paired version uses the variance of the per-sample difference $d_i = m_i^{(A)} - m_i^{(B)}$:

$$\Delta_{\min}^{(M,\text{paired})} = z_{\alpha/2} \cdot \sqrt{\frac{\mathrm{Var}(d_i)}{N}}.$$

D.2 Perplexity

For perplexity, the per-sample value is $\exp(\ell_i)$ where ℓ_i is the negative log-likelihood of the i-th example. By the Delta method, for a transformation $g(\cdot)$ applied to an estimator $\hat{\theta}$ with asymptotic variance σ_θ^2:

$$\mathrm{Var}(g(\hat{\theta})) \approx [g'(\hat{\theta})]^2 \sigma_\theta^2.$$

For perplexity, $g(\ell) = \exp(\ell)$ and $g'(\ell) = \exp(\ell)$, giving:

$$\sigma_{\text{PPL}}^2 \approx \exp(2\bar{\ell}) \cdot \frac{\sigma_\ell^2}{N},$$

where $\bar{\ell}$ is the mean log-perplexity and σ_ℓ^2 is the variance of per-token log-likelihoods. The $\Delta_{\min}$ for perplexity then follows from substituting this variance into the general formula above.

D.3 Cross-Entropy Loss

Cross-entropy loss is typically already an average, so no transformation is needed:

$$\Delta_{\min}^{(\text{CE})} = z_{\alpha/2} \cdot \sqrt{\frac{\hat{\sigma}_{\text{CE}}^2}{N}},$$

where $\hat{\sigma}_{\text{CE}}^2$ is the empirical variance of per-example losses. A conservative upper bound can be obtained by noting that for K-way classification with bounded logits, the per-example loss is bounded in $[0, \log K]$, giving $\sigma_{\text{CE}}^2 \leq (\log K)^2/4$ by Popoviciu's inequality.

D.4 BLEU and Other Discrete-Valued Metrics

For metrics like BLEU that are computed at the corpus level rather than as per-example averages, the Delta method still applies but requires care: the metric is a function of multiple sufficient statistics (n-gram counts). Let $\mathbf{c} = (c_1, \ldots, c_k)$ be the vector of corpus-level counts, with covariance Σ_c estimated via bootstrap. Then for BLEU $= B(\mathbf{c})$:

$$\mathrm{Var}(\widehat{\text{BLEU}}) \approx \nabla B(\mathbf{c})^\top \Sigma_c \nabla B(\mathbf{c}),$$

and $\Delta_{\min}^{(\text{BLEU})} = z_{\alpha/2} \cdot \sqrt{\mathrm{Var}(\widehat{\text{BLEU}})}$.

D.5 Practical Recommendations

For practitioners:

- **Pre-experimental** (no data available): Use the proportion-based $\Delta_{\min}$ (Eq. 2) as a conservative bound, since the effective variance of a continuous metric is typically lower than that of a proportion.
- **Post-hoc with per-sample metrics**: Compute the empirical variance $\hat{\sigma}^2$ from a small pilot sample and plug into the general formula above. A pilot of $N = 50$ samples is usually sufficient for a stable variance estimate.

- **For paired comparisons** (same evaluation set): Always use the paired formula, which can reduce $\Delta_{\min}$ by 20–40% when per-sample metric values are positively correlated across models.

E Detailed Literature Survey Data

The literature survey was conducted as a rapid assessment of reporting practices. The category-level summary data are available in the repository (see Code and data availability); individual paper-by-paper coding records were not retained as a structured extraction table, consistent with the survey's motivating rather than confirmatory role. The 50-paper sample can be reproduced by following the search protocol described in §6.

F Adjacent Comparison Tests

Table 7 reports the full two-proportion z-tests for adjacent checkpoint pairs at $N = 200$.

Table 7 Adjacent two-proportion z-tests (few-shot protocol, $N = 200$).

Pair	Δ	z	p
50→100	+7.0%	1.40	0.161
100→150	+1.0%	0.20	0.841
150→200	−6.5%	−1.30	0.193
200→250	+5.0%	1.00	0.317
250→300	+1.0%	0.20	0.841
300→350	−3.5%	−0.70	0.483
350→400	−1.0%	−0.20	0.841
400→450	+1.5%	0.30	0.764
450→500	−2.0%	−0.40	0.689

G Seed Replication Data

Table 8 shows the full accuracy trajectories for seeds 43 and 44.

Table 8 GSM8K accuracy (training-format prompt) across three seeds at 128 tokens (left) and 256 tokens (right).

Step	**128 tokens**			**256 tokens**		
	Seed 42	**Seed 43**	**Seed 44**	**Seed 42**	**Seed 43**	**Seed 44**
50	50.5%	51.0%	52.5%	60.0%	79.5%	77.0%
100	52.0%	52.5%	53.0%	49.5%	69.0%	73.5%
150	50.0%	54.5%	52.0%	60.5%	75.5%	74.0%
200	46.5%	50.0%	52.0%	52.0%	68.5%	63.5%
250	47.0%	54.0%	53.0%	50.0%	68.5%	71.5%
300	48.5%	50.5%	52.5%	55.0%	65.5%	67.5%
350	50.5%	49.5%	52.5%	56.0%	66.0%	65.5%
400	50.5%	50.5%	50.0%	57.0%	65.5%	65.0%
450	51.0%	53.5%	53.0%	62.0%	66.5%	66.5%
500	49.5%	52.0%	51.5%	57.5%	67.0%	66.5%
Range	5.5 pp	5.0 pp	3.0 pp	12.5 pp	14.0 pp	13.5 pp

H Detailed Pitfall Methodology

Additional context for each pitfall:

- **T1 (Spearman discretization):** We computed ρ over a sliding window of $n = 3$ steps at each of 10 checkpoint positions. The $n = 3$ window was chosen arbitrarily in early analysis; $n = 6$ was adopted after noticing that the extreme $\rho = -1$ values always coincided with exactly three-point windows.
- **T2 (Synthetic fallback):** The config file held_out_path used a path without file extension while the actual file had a .json extension. Python's json.load raised FileNotFoundError, caught by a generic exception handler that called a development-only synthetic function.
- **T3 (Definition ambiguity):** proxy_main was defined as "the primary proxy metric" without specifying which of three candidates (proxy_loss, proxy_win_rate, or a combined score) it resolved to. The resolution rule depended on initialization order.
- **T4 (Protocol mismatch):** The 7% accuracy was from prompts like "Question: ...\nAnswer:". With three in-context examples demonstrating the #### <answer> format, accuracy rose to 47–55%.
- **T5 (Code sandbox):** The sandbox environment was a minimal Python install without sklearn, pandas, or numpy. Generated code using these libraries crashed silently. Output parsers expected a specific <function> marker that only appeared in one of three generation formats.
- **T6 (Post-hoc pattern):** The initial observation (GSM8K peak at step 350 then decline) came from the same misconfigured pipeline in T4. The replication criteria, a pre-registered clean run with proper few-shot prompting, showed no such pattern.
- **T7 (Decoding sensitivity):** Evaluations were run with temperature 0.0 (greedy decoding). Switching to temperature 0.2 with 5 random seeds changed individual checkpoint accuracies by 2–3 pp, altering the ranking in 3/10 positions.
- **T8 (Metric versioning):** The evaluation script went through 5 revisions over the project timeline. Revision 2 $\rightarrow$ 3 changed the GSM8K answer extraction regex, shifting all checkpoint accuracies by +1.2 pp on average.
- **T9 (Few-shot sensitivity):** Three fixed exemplars were selected from the GSM8K training set. Three different orderings produced per-checkpoint accuracy differences of 1.5–4.0 pp, with the best model changing between orderings in 2/3 permutations.

Statements and Declarations

Funding. The author received no financial support for this work.
Competing interests. The author declares no competing interests.
Author contributions. Y.W. is the sole author and is responsible for all aspects of the work, including the literature survey coding described in §6.
Code and data availability. All code and data supporting the findings of this study are openly available at https://github.com/wyp629929/A-Resolution-Framework-for-Finite-Sample-Model-Selection. The repository contains a command-line implementation of the $\Delta_{\min}$ diagnostic in both its independent and paired forms, the per-checkpoint accuracy records underlying Tables 1, 8 and 3, the bootstrap and Monte Carlo scripts used for Figures 1–3, the MNIST random forest experiment, the per-sample cross-entropy loss extraction of §5.7, and the literature survey summary data of §6 including category-level aggregates. Model checkpoints are not redistributed; the training configurations required to reproduce them are included.

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